

Blockchain-Based Food Supply Chain Provenance System

Problem Statement

Food moves through many hands — from farmers and transporters to distributors and retailers — before it reaches consumers. Because of this, it's often hard to verify where the food came from, how it was handled, or whether it's authentic. Problems like food fraud, false labeling, and tampering damage public trust and can lead to financial losses or even health risks.

Traditional tracking systems use centralized databases, which can be changed or manipulated and don't provide full transparency for everyone involved.

This project aims to build a blockchain-based tracking system that records every step of a food product's journey — from production to consumption — on a secure and unchangeable digital ledger. This will let all participants register, check, and transfer product information transparently, creating accountability and trust across the entire supply chain.

Motivation

People today want to know where their food comes from and whether it's safe. Regulators also need reliable proof that food meets safety and quality standards. Blockchain technology can make this possible because it's decentralized (no single party controls it) and immutable (data cannot be changed once recorded).

By using a blockchain-based system for food tracking:

- Consumers can easily check a product's source and journey through a QR code or online dashboard.
- Farmers, suppliers, and retailers can reduce paperwork and human errors by using automated smart contracts.
- The system helps prevent fraud, reduces disputes, and ensures transparency for everyone involved.

Literature Review

Previous studies have demonstrated that blockchain technology can significantly enhance transparency and reliability in supply chains. IBM Food Trust [1] pioneered one of the earliest large-scale implementations in 2018, using blockchain to trace food items such as lettuce and mangoes, thereby enabling rapid identification of contamination sources. Tian [2] proposed an integrated system combining blockchain and the Internet of Things (IoT) to improve food safety through real-time tracking of environmental factors like temperature and transportation data. Building on these technological foundations, Kouhizadeh and Sarkis [3] discussed how blockchain facilitates ethical and sustainable sourcing practices by improving traceability and accountability within food supply chains. Similarly, Saberi et al. [4] examined the broader role of blockchain in promoting environmentally responsible and transparent supply chain management practices.

While these projects showcase the potential of blockchain, many depend on complex IoT infrastructures or costly implementation environments. In contrast, our project presents a **simplified Ethereum-based prototype** that demonstrates the core principles of blockchain — including smart contracts, transparency, and role-based access control — without requiring advanced hardware or large-scale deployment.

Proposed Architecture

The system will be built on the **Ethereum test network** using **Solidity** for smart contract development.

Stakeholders

- **Farmer/Producer:** Registers the food batch with details such as ID, date, and origin.
- **Distributor:** Records transport and delivery information.
- **Retailer:** Confirms product receipt and updates status.
- **Regulator (optional):** Checks certification and compliance data.
- **Consumer:** Scans a QR code to view the verified history of the product.

Smart Contract Design

- **Data Structure:** Stores product ID, owner, status, timestamps, and extra information.
- **Functions:** registerProduct(), transferOwnership(), updateStatus(), and verifyProduct().
- **Events:** ProductCreated, OwnershipTransferred, and StatusUpdated help track changes.
- **Access Control:** Solidity modifiers will make sure only authorized users can perform certain actions.

Front-End Interface

A **simple web dashboard** (built with React or HTML + Web3.js) will allow users to add or view product data on the blockchain.

Consumers can scan a **QR code** to see the full product history and verify authenticity using blockchain records.

Future Improvements

Later, the project could add **off-chain storage** (like IPFS) to hold certificates or IoT data such as temperature or transport conditions.

References

1. Tian, F. (2017). *A supply chain traceability system for food safety based on HACCP, blockchain & Internet of Things*. 14th International Conference on Service Systems and Service Management.
2. Kouhizadeh, M., & Sarkis, J. (2018). *Blockchain practices, potentials, and perspectives in sustainable supply chain management*. Sustainability, 10(10), 3652.
3. IBM Food Trust. (2018). *Blockchain-enabled transparency for the food supply chain*. IBM Research White Paper.
4. Saberi, S., Kouhizadeh, M., Sarkis, J., & Shen, L. (2019). *Blockchain technology and its relationships to sustainable supply chain management*. International Journal of Production Research, 57(7), 2117–2135.

Team Contract

Roles

- **Smart Contract Developer:** Builds and tests Solidity contracts for registering products, tracking transfers, and updating product status.
- **Front-End Developer:** Creates the user dashboard and connects it to the blockchain using Web3.js.
- **Research Lead:** Handles research, documentation, and analysis of the project's performance and ethical aspects.
- **Project Manager:** Organizes meetings, tracks progress, and ensures that the project meets deadlines and submission requirements.

Expectations

- Every team member contributes equally to coding, research, and writing.
- Weekly progress reports will be shared during team meetings.
- All project files will be stored in a shared GitHub repository with version control.
- Major updates and code changes must be reviewed by another team member before merging.

Meeting Frequency

- **Weekly Meetings:** One-hour sessions every Friday to discuss progress and assign new tasks.
- **Extra Sessions:** Additional meetings will be scheduled when needed for debugging or integration.
- **Communication:** The team will stay in touch regularly through Discord or Slack.